CareerCompass AI

Skill-based Career Recommender, Resume Builder & Interview Coach

Concept Note

IBM SkillsBuild - Masterclass 5: Creating Real-Life Projects
Aligned with UN SDG 8: Decent Work and Economic Growth

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1. Introduction

Every year lakhs of graduates enter the job market, but many - especially in Tier-2 and Tier-3 cities - do not know which career suits their skills, what they still need to learn, or how to present themselves to employers. Professional career counselling is costly and rarely available. CareerCompass AI is a web application that gives every student a free, personalised career guide: it matches skills to careers, builds a learning roadmap for the gaps, creates a scored ATS-friendly resume and lets users practise interview answers.

2. Problem Statement

"Many skilled individuals in Tier-2/3 cities face employment challenges because they lack guidance on suitable careers, the skills they are missing, and job-ready resumes. This AI project suggests personalised career paths based on user skills and location, and helps users become job-ready."

- No easy way to know which careers match one's current skills.
- Skill gaps are unclear and free learning material is scattered.
- Resumes lack action verbs, numbers and structure, so they are rejected.
- Little interview practice, leading to low confidence.
- Job availability differs sharply between metro and small-town locations.

3. Objectives

- Rank 15 careers from 42 skills using a weighted match that also considers city tier and remote work.
- Generate a skill-gap roadmap with estimated weeks and free resources, with progress tracking.
- Build and score resumes on 7 criteria and export a clean PDF.
- Provide HR and role-specific mock interview questions with instant keyword-based feedback.
- Offer a secure personal dashboard for plans, resumes and practice history.

4. Alignment with SDG 8

The project supports **SDG 8 - Decent Work and Economic Growth**: Target 8.5 (full and productive employment), Target 8.6 (reducing youth not in employment, education or training) and Target 8.b (youth employment strategy). It directly implements the masterclass sample ideas: a resume builder and career guidance agent, and a job prediction agent based on skills and location.

5. Proposed Solution - Modules

Module	What it does	How it works
Career Matcher	Top career matches with salary range, demand, local job availability and remote flag	Score = 85% weighted skill match + 15% location availability (tier-wise, or full for remote-friendly roles)
Career Explorer	Browse all careers by category with key skills	Curated knowledge base
Skill-gap Roadmap	Ordered list of missing skills, priority, weeks and free course	Must-have skills first; save as plan and tick off progress
Resume Builder	Structured resume, preview, 0-100 score with tips, PDF export	Checks contact, summary length, skills, action verbs and numbers
Mock Interview	4 HR + 2 technical questions per career, score and feedback	Keyword coverage (60), answer length (20), example usage (20)
Dashboard	Plans with progress bars, resumes, interview history	SQLite database; per-user isolation

6. System Architecture

The application has three tiers: a responsive HTML/CSS front end (Jinja2), a Flask back end with a separate decision engine (engine.py) and knowledge base (data.py), and an SQLite database for users, plans, resumes and interview attempts. PDFs are created with ReportLab. The system runs fully offline, so personal data stays with the user.

Layer	Technology	Responsibility
Presentation	HTML5, CSS3, Jinja2	Forms, results, resume preview
Application	Python 3, Flask	Routing, validation, sessions, CSRF protection
Logic	engine.py, data.py	Matching, roadmap, resume and interview scoring
Data	SQLite	Users, plans, resumes, interviews
Reporting	ReportLab	Resume PDF

7. Methodology

- **Requirement analysis** from the problem statement and review of existing alternatives.
- **Knowledge-base design** - skills, career weights, salary ranges, tier-wise availability, learning resources, interview questions.
- **Modular development** - decision logic separated from the web layer.
- **Testing** - 20 automated tests covering every function, route, validation rule and security check (all passing).
- **Deployment** - one-command local run; can be hosted on any cloud platform.

8. Security, Privacy and Ethics

- Salted password hashing, CSRF tokens on every form, output escaping against XSS.
- Users can access only their own plans, resumes and history.
- Transparent scoring: users see exactly why a career or resume received its score.
- Salaries are labelled as indicative; the tool guides but does not guarantee employment.

9. Expected Impact

- Helps students choose careers with confidence and reduces mismatch.
- Gives rural and Tier-2/3 youth access to guidance that was only available in metros.
- Raises resume quality and interview readiness, improving employability.
- Supports colleges' placement cells with a scalable, free tool.

10. Limitations and Future Scope

- Data is curated and indicative; it should be refreshed regularly from real job-market sources.
- Future: live job listings API, resume upload and parsing, AI/LLM-based interview evaluation, regional languages, recruiter portal and mobile app.

11. Conclusion

CareerCompass AI shows how a well-structured, fully tested application can turn career confusion into a clear, actionable plan. It is low-cost, scalable and directly contributes to SDG 8, making it a strong real-life project with startup potential.